POLYGLOT WORKSHOP

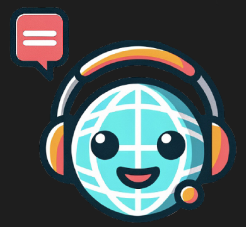
SETTING UP

Tutor
Aniket Sen

TRA SUSTAINABLE FUTURES

UNIVERSITÄT BONN

WHY WE NEED TO SET UP?



- This is a hands-on workshop $\longrightarrow$ we'll be coding a bunch of stuff
- Working with code $\longrightarrow$ version control nightmare

WHAT I MEAN?

```
import torch

scaler = torch.cuda.amp.GradScaler()
```

works fine for `PyTorch >= 1.6`

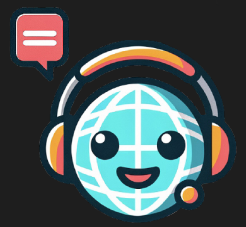
but for `PyTorch >= 2.4` we get

```
ModuleAttributeError: module 'torch.cuda.amp' has no attribute 'GradScaler'
```

WHY?

because in `PyTorch 2.4`, `GradScaler` was relocated to `torch.amp.Gradscaler`

WHY WE NEED TO SET UP?



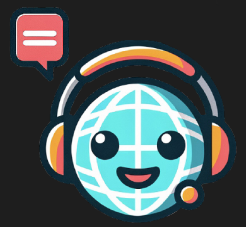
- We will also be working together → so we need to be on the same page
- So that all our bugs bug us equally 😄

HOW CAN WE ALL BE ON THE SAME PAGE?

- We all work on the same environment
- We all work with the same software stack

IT ALSO MAKES LIFE MUCH EASIER FOR US!! 😅

HOW CAN WE BE ON THE SAME ENVIRONMENT?



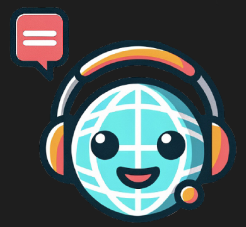
- We use *Marvin*!!

Marvin is a State of the Art (for 2022) Tier 3 HPC cluster at the University of Bonn

- We have created temporary accounts on Marvin for all of you
  - you should have received an email (please say now if you haven't)
- We have also reserved a few nodes for the workshop
- We will be doing everything on Marvin remotely

A HUGE THANKS TO THE HPC TEAM OF UNI BONN!

HOW DO WE WORK REMOTELY ON MARVIN?



- Using *SSH* (Secure Shell)
- For that we need to set up a SSH key pair for you to access Marvin securely

Open `Terminal` on your laptop and type the following command

```
ssh-keygen -t ed25519
```

you will be asked to enter a file name to save the key

```
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/<username>/.ssh/id_ed25519):
```

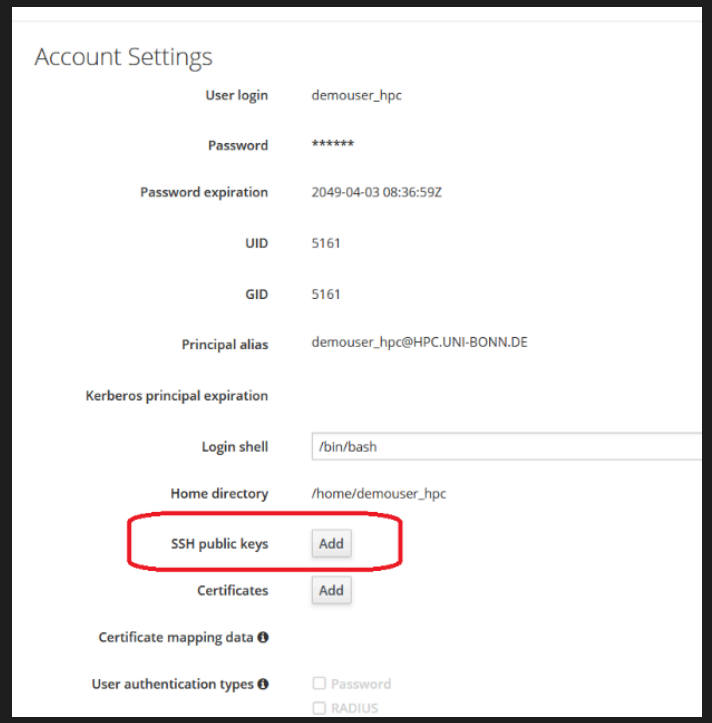
you will also be asked to enter a passphrase

```
Enter passphrase for "/home/<username>/.ssh/id_ed25519" (empty for no passphrase):
```

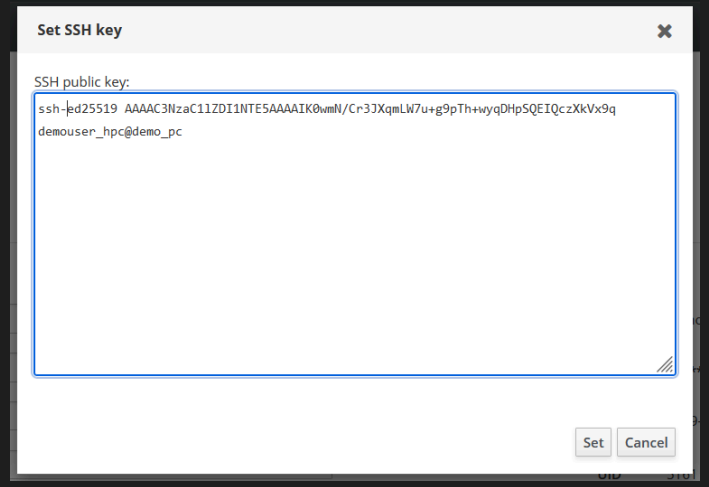
IMPORTANT!! Please enter a strong passphrase and remember it, otherwise you won't be able to access Marvin

PROVIDING THE PUBLIC KEY TO MARVIN

- Go to the Marvin user portal: https://freeipa.hpc.uni-bonn.de/
- Log in with the credentials you received via email from HPC@UniBonn

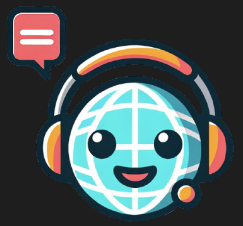


- Click on `SSH public keys -> Add` on the right side of the page
- Copy the content of your public key file (e.g., `~/.ssh/id_ed25519.pub`) and paste it into the `Public key` field

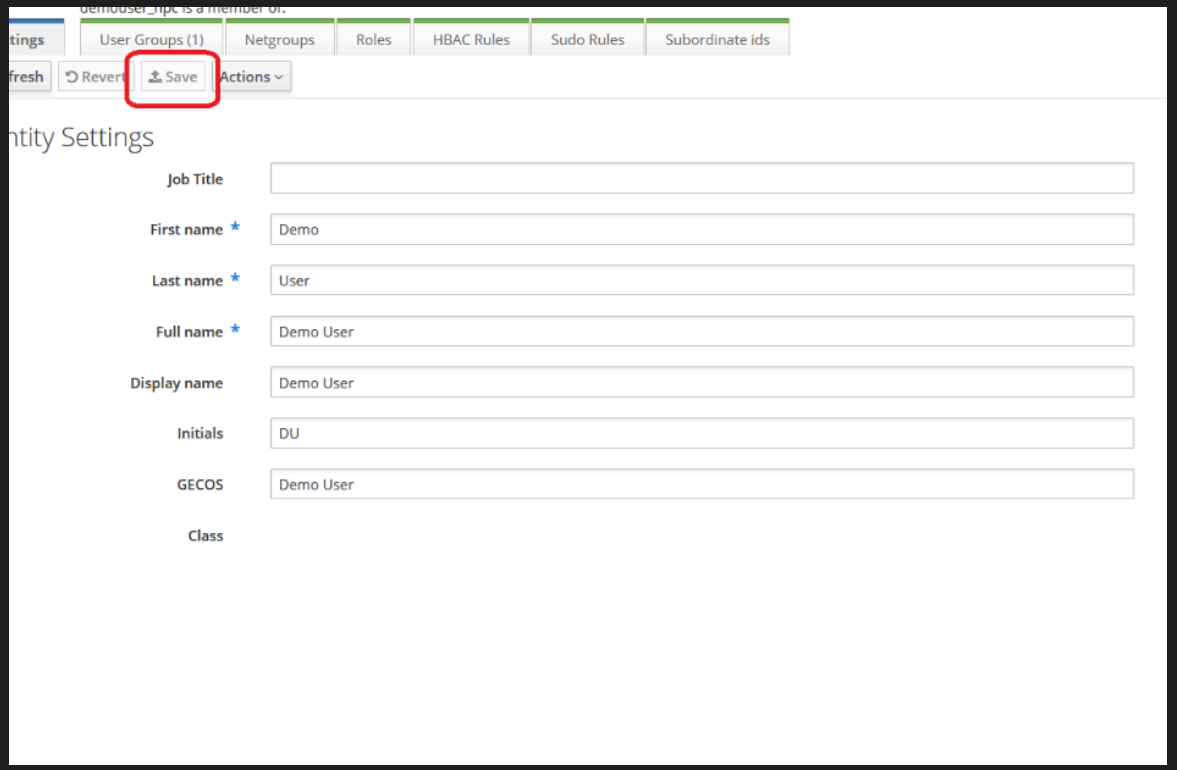


- Click on `Set` to save the public key

PROVIDING THE PUBLIC KEY TO MARVIN



- Go to the Marvin user portal: https://freeipa.hpc.uni-bonn.de/
- Log in with the credentials you received via email from HPC@UniBonn



And finally, click `Save` in the top left bar to save the changes to your account

If something went wrong, please tell us now!

SET UP AN SSH CONFIG FILE

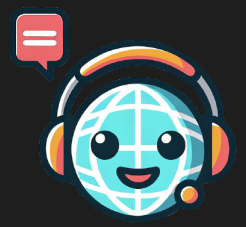
- To make it easier to connect to Marvin, we will set up an SSH config file on your laptop
- This will allow us to connect to Marvin using a simple command like `ssh marvin` instead of `ssh -i /path/to/your/private/key <username>@marvin.hpc.uni-bonn.de`

Create a file named `config` in the `~/.ssh/` directory (if it doesn't already exist)

And add the following content to the `config` file

```
Host marvin
  HostName marvin.hpc.uni-bonn.de
  User <username>
  IdentityFile /path/to/your/private/key (e.g., ~/.ssh/id_ed25519)
  IdentitiesOnly yes
```

LET'S TEST IT OUT!



Open a new terminal and type the following command

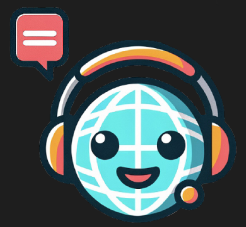
```
ssh marvin
```

If everything is set up correctly, you will be greeted with this

```
        Welcome to the marvin-cluster!
        For more information check our wiki: https://wiki.hpc.uni-bonn.de

        Do not hesitate to contact marvin-support@hpc.uni-bonn.de if you
        have any questions or issues.

                                *** IMPORTANT ***
        There is an issue with the lustre-filesystem. Some users are writing
        absurd amount of small files to the MetaDataTargets, blocking them for
        everyone else. Please make sure to understand lustre and use it
        properly.
        The issues regarding the home filesystem are resolved.
        Please note the changes regarding ssh-login after our
        downtime on 2024-09-25. Please refer to our wiki for further information
        here https://wiki.hpc.uni-bonn.de/gaining_access#connecting-via-ssh


Last login: Fri Mar  6 10:25:11 2026 from [redacted]

Your Lustre Quota Usage:
Disk quotas for usr asen_hpc (uid 225605):
     Filesystem    used   quota   limit   grace   files   quota   limit   grace
        /lustre    452k      0k      0k       -      17       0       0       -
[asen_hpc@login02 ~]$
```

Let us know now if you do not see this welcome message after running `ssh marvin`

LET'S CREATE A WORKSPACE FOR YOU!



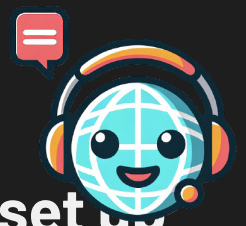
In the terminal where you are connected to Marvin, run the following command to create a workspace for you

```
wget -O create_workspace_polyglot.sh https://tinyurl.com/polyglot-st-cr \
  && chmod +x create_workspace_polyglot.sh \
  && ./create_workspace_polyglot.sh
```

This will create a workspace for you in the `/lustre/scratch/data/<username>-polyglot` directory

Next we will set up some environment variables that will be useful for us later on

SETTING UP ENVIRONMENT VARIABLES



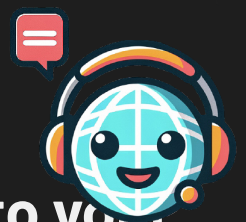
In the terminal where you are connected to Marvin, run the following command to set up the environment variables

```
wget -O set_env_vars_polyglot.sh https://tinyurl.com/polyglot-st-env \
  && source set_env_vars_polyglot.sh
```

This will set up the following environment variables for you:

- `WS_DIR` : the directory of your workspace (e.g., `/lustre/scratch/data/<username>-polyglot`)
- `cdws` : a shortcut command to change to your workspace directory (i.e., `cd $WS_DIR`)
- `WS_COMMON` : the directory of the common files that we will be using in the workshop (`/lustre/scratch/data/nklugeco_hpc-workshop`)
- `sbatch_poly` : a shortcut command to submit a job to the cluster using the correct reservation and account (`sbatch --reservation=polyglot_[mon,tue...] --account=tmp_polyglot`)

SETTING UP ENVIRONMENT VARIABLES



To make the environment variables available throughout the workshop, we add it to your `.bashrc` (this is a script that runs every time you log in to Marvin)

```
echo "source ~/set_env_vars_polyglot.sh" >> ~/.bashrc
```

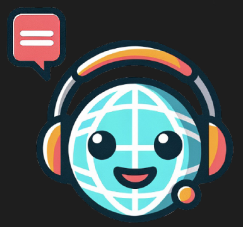
Its completely understandable if you don't want to do this (especially if you already use Marvin regularly)

In that case, you need to run the following every time you start a new session on Marvin

```
source ~/set_env_vars_polyglot.sh
```

That was a lot of setup (and working with a terminal can be intimidating), so let's take a break!!

LET'S MAKE THINGS SLIGHTLY EASIER FOR YOU!



Working with the terminal can be intimidating, especially if you are not used to it

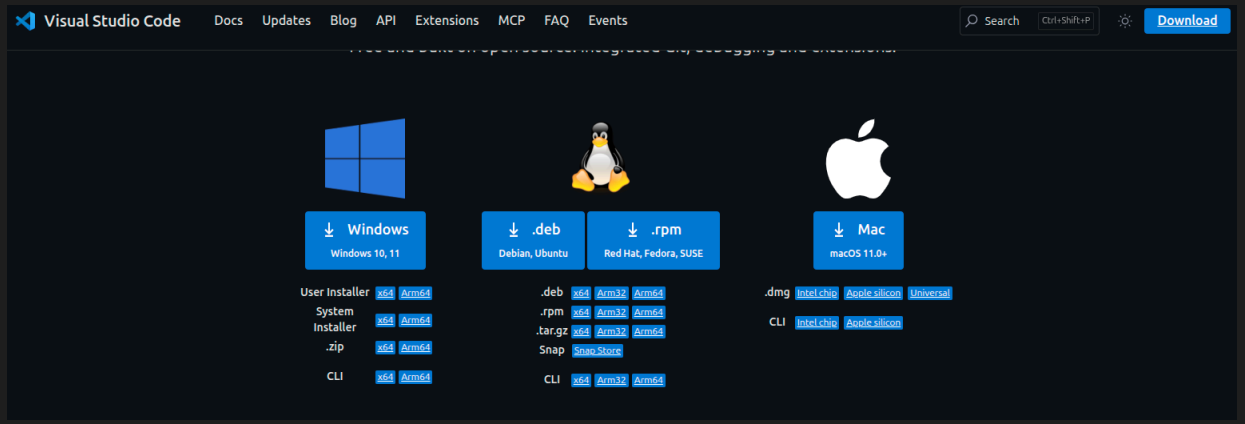
To make things easier for you, we would do everything through `VSCode`

This is completely optional, you can still follow along using the terminal if you prefer

You can download `VSCode` from the official website:

https://code.visualstudio.com/download

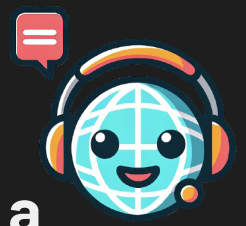
Choose the option that suits your system



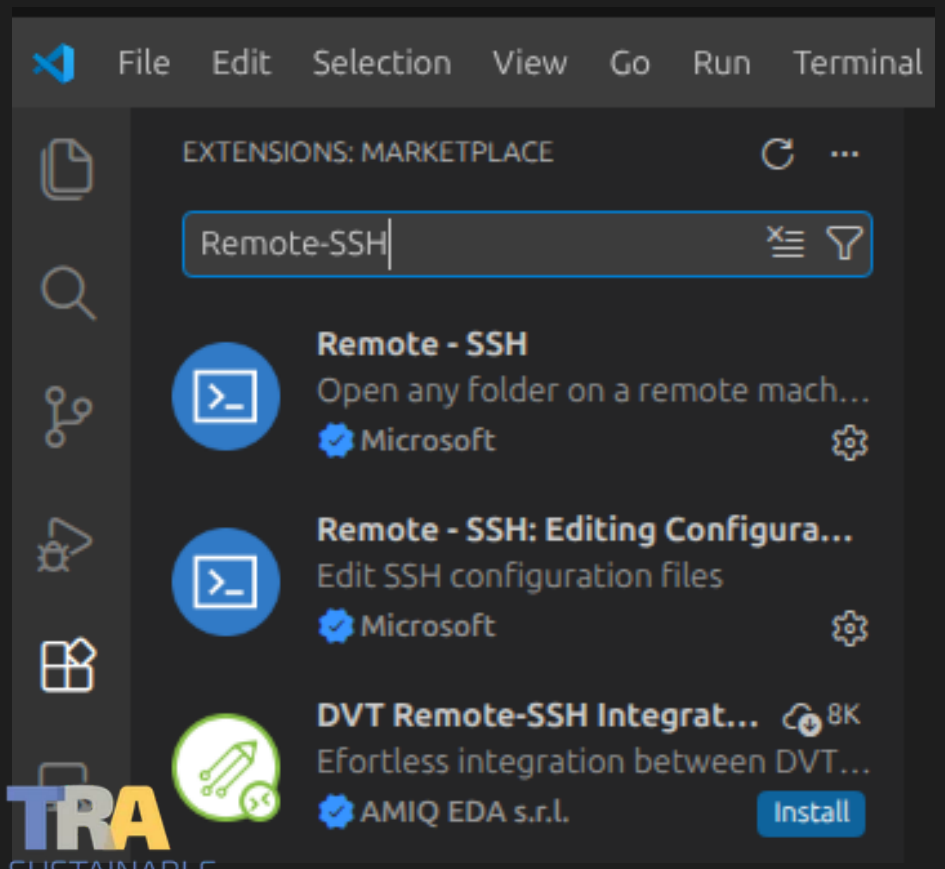
Feel free to ask us for help if you are not sure which one to choose or if you have any issues installing `VSCode`

TRA SUSTAINABLE FUTURES
UNIVERSITÄT BONN

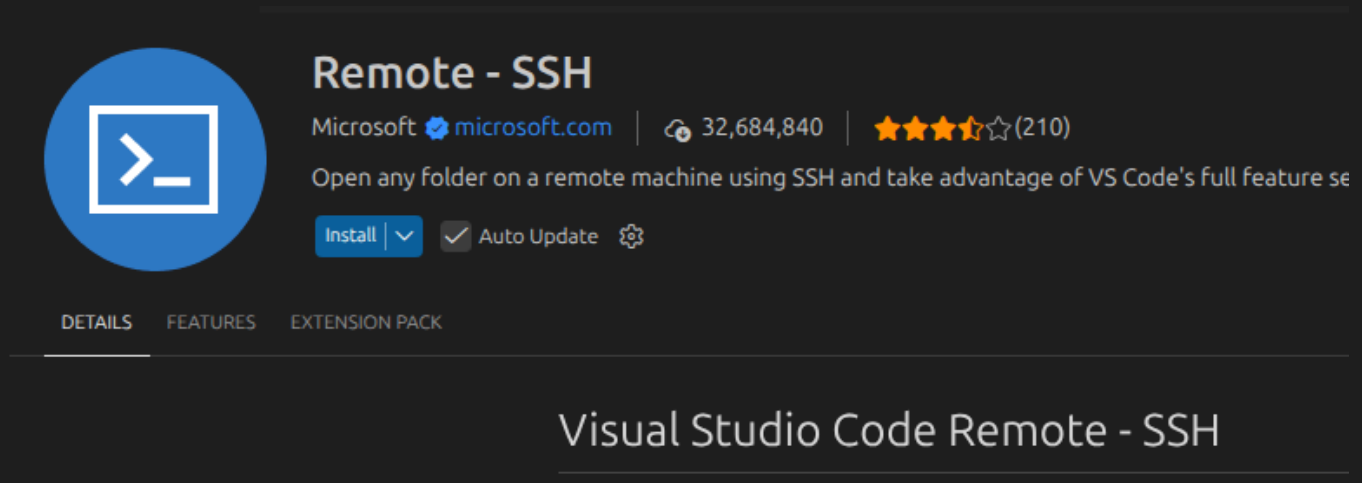
REMOTE DEVELOPMENT WITH VSCODE



`VSCode` has a great extension called `Remote - SSH` that allows you to connect to a remote server (like Marvin) and work with the files on that server as if they were on your local machine

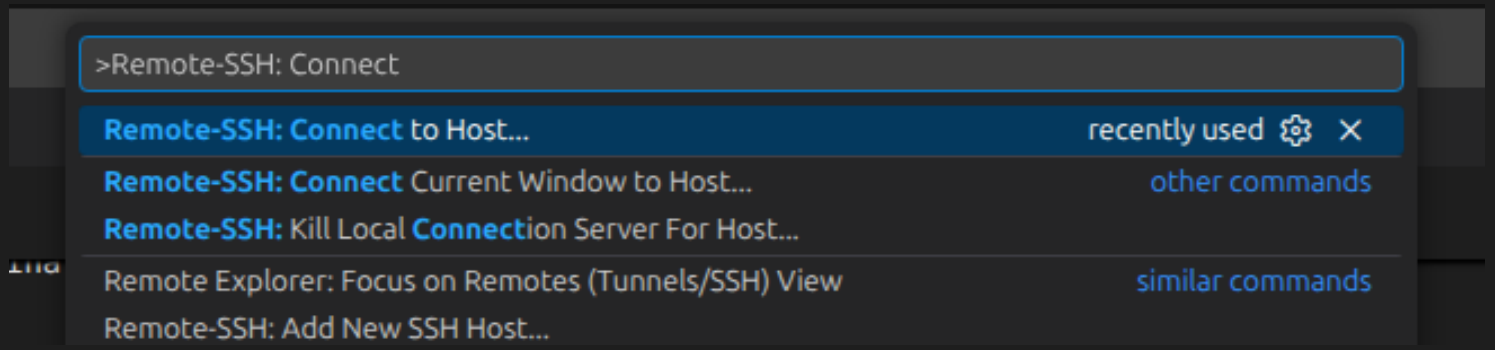


Got to the extensions tab in `VSCode` and search for `Remote - SSH` and install it

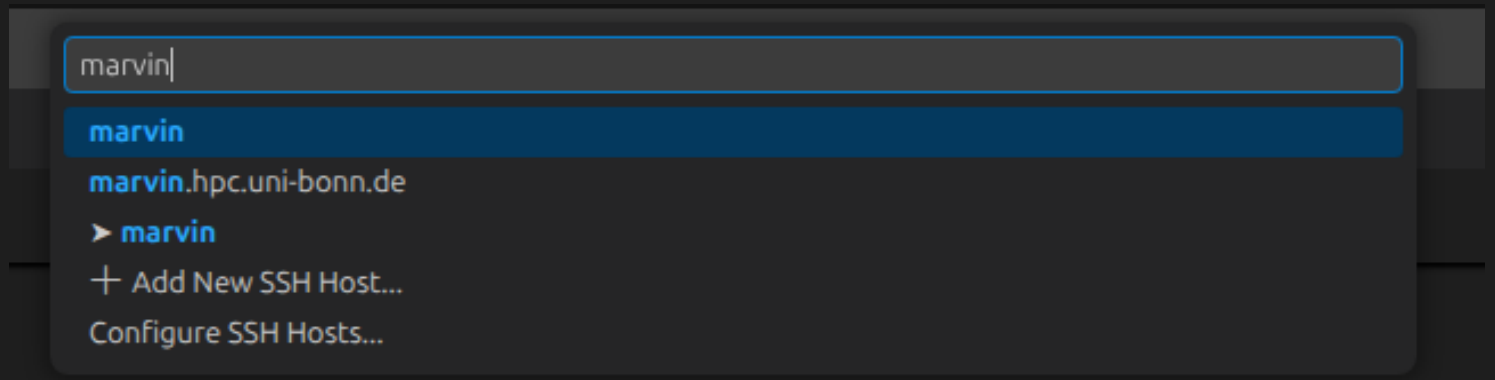


CONNECTING TO MARVIN WITH VSCODE

On `VSCode`, press `F1` or `Ctrl + Shift + P` to open the command palette and type



Select `marvin` from the list of hosts (if you have many)

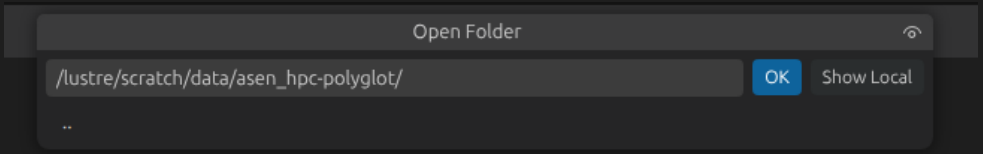


If all your previous steps were successful, you should be now running a `VSCode` server on Marvin

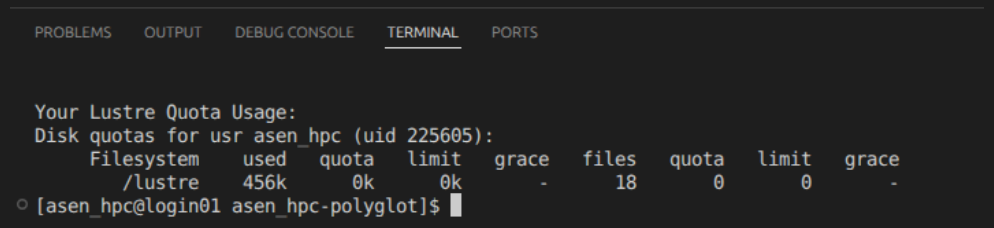
SETTING UP VSCODE FOR REMOTE DEVELOPMENT

Now that we are connected to Marvin with `VSCode`, we open the workspace directory that we created earlier

Press `Ctrl + k` followed by `Ctrl + o` (or `File -> Open Folder`) and navigate to your workspace directory



We can also open up a terminal in `VSCode` at this directory by going to `Terminal -> New Terminal` (or pressing `Ctrl + Shift + ` `)



PERFECT!! NOW WE ARE ALL WORKING WITH THE SAME ENVIRONMENT 🎉

This is going to be our main window for the rest of the workshop